V SEMESTER (PROFESSIONAL ELECTIVE-I)	L	T	P	C
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20AM5E02-COMPUTER VISION				

COURSE OBJECTIVES:
1. To introduce students the fundamentals of image formation.
2. To introduce students the major ideas, methods, and techniques of computer vision and pattern recognition.
3. To develop an appreciation for various issues in the design of computer vision and object recognition systems.
4. To provide the student with programming experience from implementing computer vision and object recognition applications.

COURSE OUTCOMES :
At the end of the course, student will be able to
CO1: Identify basic concepts, terminology, theories, models and methods in the field of computer vision[K1].
CO2: Describe known principles of feature detection and matching[K2].
CO3: Describe basic methods of computer vision related to image stitching, photography like high dynamic range imaging and blur removal[K2].
CO4: Suggest a design of a computer vision system for a 3D Recon- struction, Albedos, image based rendering views and depths.[K3].
CO5: Understanding of 3D reconstruction techniques and image-based rendering methods.

UNIT - I :
Introduction: Image Formation: Geometric Primitives and Transformation, Photometric Image Formation, Digital Camera, Image Processing: Point Operators, Linear Filtering, More Neighborhood Operators, Fourier Transforms, Pyramids and Wavelets, Geometric Transformations, Global Optimization.

UNIT - II :
Feature Detection and Matching: Points and Patches, Edges, Lines, Segmentation: Ac- tive Contours, Split and Merge, Mean Shift and Mode Finding, Normalized Cuts, Feature- Based Alignment: 2D and 3D Feature-based Alignment, Pose Estimation, Geometric In- trinsic Calibration.

UNIT - III :
Structure and Motion: Triangular, Two-frame Structure from Motion, Factorization, Bundle Adjustment, Constrained Structure and Motion, Dense Motion Estimation: Trans- lation Alignment, Parametric Motion, Spline-based Motion, Optical Flow, Layered motion

UNIT - IV :
Image Stitching: Motion Models, Global Alignment, Composing, Computational Photog- raphy: Photometric Calibration, High Dynamic Range Imaging, Super-Resolution and Blur Removal, image Matting and Compositing, Texture Analysis and Synthesis.

UNIT - V :
3D Reconstruction: Shape From X, Active Range Finding, Surface Representation, Point-based Representation, Volumetric Representation, Model-based Reconstruction, Re- covering Texture Maps and Albedos, Image- based Rendering: View Interpolation, Layered Depth Images, Light Fields and Lumigraphs, Environment Mattes, Video-based Rendering.

Text Books:
1. Richard Szeliski, Computer Vision: Algorithms and Applications, Springer-Verlag London Limited, 2011.
2. Simon J.D Prince, Computer Vision: Models, Learning and Inference, 1st Edition, 2012.

Reference Books:

1. Computer Vision - A modern approach, by D. Forsyth and J. Ponce, Prentice Hall Robot Vision, by B. K. P. Horn, McGraw-Hill.
2. Haralick & Shapiro, "Computer and Robot Vision", Vol II.3
3. Gerard Medioni and Sing Bing Kang "Emerging topics in computer vision"

NPTEL LINK:

1. https://onlinecourses.nptel.ac.in/noc22 ee48/preview